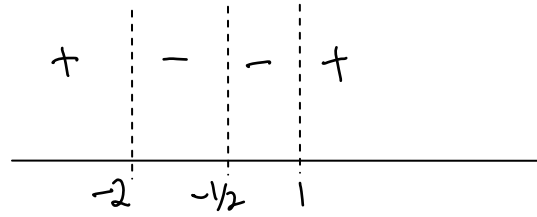


2009 Prelim H2 Maths Paper 1

1. Objective: To test awareness of inequality with repeated roots

$$\begin{aligned}\frac{9}{2-x-x^2} \leq 4 &\Rightarrow 4 - \frac{9}{2-x-x^2} \geq 0 \\ &\Rightarrow 4 + \frac{9}{x^2+x-2} \geq 0 \\ &\Rightarrow \frac{4x^2+4x+1}{x^2+x-2} \geq 0 \\ &\Rightarrow \frac{(2x+1)^2}{(x+2)(x-1)} \geq 0\end{aligned}$$



Using sign test or graphical method, $x < -2$ or $x = -\frac{1}{2}$ or $x > 1$.

2. Objective: R formula & transformation

$$y = \sqrt{3} \sin 2x + \cos 2x - 1$$

$$\text{Let } \sqrt{3} \sin 2x + \cos 2x = R \sin(2x + \alpha)$$

$$= R \sin 2x \cos \alpha + R \cos 2x \sin \alpha$$

$$\rightarrow \tan \alpha = \frac{1}{\sqrt{3}} \quad \alpha = \frac{\pi}{6}$$

$$R = 2$$

$$\therefore y = 2 \sin\left(2x + \frac{\pi}{6}\right) - 1$$

$$y = 2 \sin 2x \rightarrow y_1 = 2 \sin\left(2x + \frac{\pi}{6}\right) \rightarrow y_2 = 2 \sin\left(2x + \frac{\pi}{6}\right) - 1$$

: translation by $-\frac{\pi}{6}$ units in dir of x -axis

: translation by -1 in dir of y -axis

3. Objective: De Moivre's theorem

$$(i) \quad 1-i = \sqrt{2}e^{-\frac{\pi}{4}i}$$

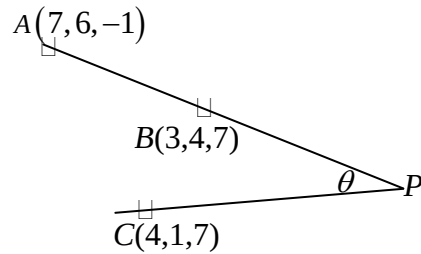
$$3(ii) \quad S_{11} = \frac{z^{11}-1}{z-1}$$

$$\begin{aligned}
&= \frac{2^{\frac{11}{2}} e^{-\frac{11\pi}{4}i} - 1}{\sqrt{2} e^{-\frac{\pi}{4}i} - 1} \\
&= \frac{2^{\frac{11}{2}} e^{-\frac{3\pi}{4}i} - 1}{\sqrt{2} e^{-\frac{\pi}{4}i} - 1} \\
&= \frac{2^{\frac{11}{2}} [\cos(-\frac{3}{4}\pi) + i \sin(-\frac{3}{4}\pi)] - 1}{\sqrt{2} [\cos(-\frac{\pi}{4}) + i \sin(-\frac{\pi}{4})] - 1} \\
&= \frac{2^{\frac{11}{2}} \left[-\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}} \right] - 1}{\sqrt{2} \left[-\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}} \right] - 1} \\
&= 32 - 33i
\end{aligned}$$

4. Objective: Vectors application

(i) $2\mathbf{p} + \begin{pmatrix} 7 \\ 6 \\ -1 \end{pmatrix} = 3 \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$

$\mathbf{p} = \begin{pmatrix} 1 \\ 3 \\ 8 \end{pmatrix}$



(ii) $\overrightarrow{AP} \cdot \overrightarrow{CP} = |\overrightarrow{AP}| |\overrightarrow{CP}| \cos \theta$

$$\begin{pmatrix} -6 \\ -3 \\ 9 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} = \sqrt{126} \sqrt{14} \cos \theta$$

$$\cos \theta = \frac{21}{\sqrt{126} \sqrt{14}}$$

$$\theta = 60^\circ$$

(iii) $\text{Area} = \frac{1}{2} |\overrightarrow{AP}| |\overrightarrow{CP}| \sin \theta$

$$\begin{aligned}
&= \frac{1}{2} \sqrt{126} \sqrt{14} \frac{\sqrt{3}}{2} \\
&= \frac{21\sqrt{3}}{2}
\end{aligned}$$

OR $\text{Area} = \frac{1}{2} |\overrightarrow{AP} \times \overrightarrow{CP}| = \frac{1}{2} \left| \begin{pmatrix} -6 \\ -3 \\ 9 \end{pmatrix} \times \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} \right| = \frac{21\sqrt{3}}{2}$

5. Objective: Mathematical Induction involving recurrence relation

$$(i) \quad u_1 = \frac{1}{2}, \quad 2u_2 = u_1 + 2^0 \Rightarrow u_2 = \frac{3}{4}$$

$$2u_3 = u_2 + 2^{-1} \Rightarrow u_3 = \frac{5}{8}, \quad 2u_4 = u_3 + 2^{-2} \Rightarrow u_4 = \frac{7}{16}$$

$$(ii) \quad u_n = \frac{2n-1}{2^n}$$

$$(iii) \quad \text{Let } P_n \text{ be the statement } u_n = \frac{2n-1}{2^n}, \quad n \in \mathbb{N}^+$$

$$\text{When } n = 1, \text{ LHS} = u_1 = \frac{1}{2}, \text{ RHS} = \frac{2(1)-1}{2} = \frac{1}{2}$$

Hence P_1 is true

$$\text{Assume } P_k \text{ is true of some } k \in \mathbb{N}^+, \text{ i.e. } u_k = \frac{2k-1}{2^k}$$

$$\text{To prove that } P_{k+1} \text{ is also true, i.e. } u_{k+1} = \frac{2(k+1)-1}{2^{k+1}} = \frac{2k+1}{2^{k+1}}$$

$$\begin{aligned} \text{LHS} = u_{k+1} &= \frac{1}{2} [u_k + 2^{1-k}] \\ &= \frac{1}{2} \left[\frac{2k-1}{2^k} + \frac{2}{2^k} \right] = \frac{2k+1}{2^{k+1}} \\ &= \text{RHS} \end{aligned}$$

Hence P_{k+1} is true

Since P_1 is true, and P_k is true $\Rightarrow P_{k+1}$ is also true, hence by mathematical induction, P_n is true for $n \in \mathbb{N}^+$.

6. Objective: Max Min application

$$AD = 2\sqrt{r^2 - x^2}$$

$$\rightarrow P = 2x + 2 \left[2\sqrt{r^2 - x^2} \right]$$

$$= 2x + 4\sqrt{r^2 - x^2}$$

$$\frac{dP}{dx} = 2 - 4x(r^2 - x^2)^{-\frac{1}{2}}$$

$$\frac{dP}{dx} = 0 \rightarrow 2 - 4x(r^2 - x^2)^{-\frac{1}{2}} = 0$$

$$\sqrt{r^2 - x^2} = 2x$$

$$\frac{AB}{BC} = \frac{1}{k}$$

$$\rightarrow \frac{x}{2\sqrt{r^2 - x^2}} = \frac{1}{k}$$

$$\frac{1}{k} = \frac{1}{4} \rightarrow k=4$$

7. **Objective: Partial fractions, Summation of series & Method of difference**

(i) Let $\frac{1}{r(r+2)} = \frac{A}{r} + \frac{B}{r+2}$, $A = \frac{1}{2}, B = -\frac{1}{2}$

$$\frac{1}{r(r+2)} = \frac{1}{2} \left(\frac{1}{r} - \frac{1}{r+2} \right)$$

7(ii) $\sum_{r=1}^n \frac{1}{r(r+2)} = \frac{1}{2} \sum_{r=1}^n \left(\frac{1}{r} - \frac{1}{r+2} \right)$

$$= \frac{1}{2} \left[\begin{array}{ccc} \frac{1}{1} & - & \frac{1}{3} \\ \frac{1}{2} & - & \frac{1}{4} \\ \cancel{\frac{1}{3}} & - & \cancel{\frac{1}{5}} \\ & & \\ \cancel{\frac{1}{n-2}} & - & \cancel{\frac{1}{n}} \\ \cancel{\frac{1}{n-1}} & - & \frac{1}{n+1} \\ \cancel{\frac{1}{n}} & - & \frac{1}{n+2} \end{array} \right]$$

$$= \frac{1}{2} \left(\frac{3}{2} - \frac{1}{n+1} - \frac{1}{n+2} \right)$$

$$= \frac{3}{4} - \frac{1}{2} \left(\frac{1}{n+1} + \frac{1}{n+2} \right)$$

(iii) Series is convergent since as $n \rightarrow \infty, \frac{1}{n+1} \rightarrow 0, \frac{1}{n+2} \rightarrow 0$, therefore the series is

convergent. Sum to infinity = $\frac{3}{4}$

(iv) $\frac{1}{48} + \frac{1}{63} + \frac{1}{80} + \dots + \frac{1}{399} = \sum_{r=6}^{19} \frac{1}{r(r+2)}$

$$= \sum_{r=1}^{19} \frac{1}{r(r+2)} - \sum_{r=1}^5 \frac{1}{r(r+2)}$$

$$= \frac{1}{2} \left(\frac{1}{6} + \frac{1}{7} - \frac{1}{20} - \frac{1}{21} \right) = \frac{89}{840}$$

8. Objective: Find equation of tangent and normal with coordinate geometry and use of double angle formula

$$x = \sin^3 \theta \Rightarrow \frac{dx}{d\theta} = 3\sin^2 \theta \cos \theta, \quad y = \cos^3 \theta \Rightarrow \frac{dy}{d\theta} = -3\cos^2 \theta \sin \theta,$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{-3\cos^2 \theta \sin \theta}{3\sin^2 \theta \cos \theta} = -\cot \theta$$

At P , $\theta = p$

$$\text{Equation of tangent: } y - \cos^3 p = -\cot p (x - \sin^3 p)$$

At T ,

$$0 - \cos^3 p = -\cot p (x - \sin^3 p)$$

$$\Rightarrow \frac{-\cos^3 p}{-\cot p} = x - \sin^3 p$$

$$\Rightarrow x = \cos^2 p \sin p + \sin^3 p = \sin p (\cos^2 p + \sin^2 p) = \sin p$$

Coordinates of $T = (\sin p, 0)$

At N ,

$$y - \cos^3 p = -\cot p (0 - \sin^3 p)$$

$$\Rightarrow y = \cot p \sin^3 p + \cos^3 p$$

$$\Rightarrow y = \cos p \sin^2 p + \cos^3 p = \cos p (\sin^2 p + \cos^2 p) = \cos p$$

Coordinates of $N = (0, \cos p)$

$$\text{Area of } OTN = \frac{1}{2} (\sin p) (\cos p) = \frac{1}{2} \left(\frac{1}{2} \sin (2p) \right) = \frac{1}{4} \sin (2p) \text{ (shown)}$$

9. Objective: Use of sum of area of rectangles to estimate area under graph

$$\begin{aligned} \text{Total area of four rectangles} &= \frac{1}{4} \left[\frac{2}{1+\frac{5}{4}} + \frac{2}{1+\frac{6}{4}} + \frac{2}{1+\frac{7}{4}} + \frac{2}{1+2} \right] \\ &= \frac{1}{4} \left[\frac{2}{1+\frac{5}{4}} + \frac{2}{1+\frac{6}{4}} + \frac{2}{1+\frac{7}{4}} + \frac{2}{1+\frac{8}{4}} \right] \\ &= \frac{1}{4} \left[\frac{2(4)}{9} + \frac{2(4)}{10} + \frac{2(4)}{11} + \frac{2(4)}{12} \right] \\ &= \frac{2}{9} + \frac{2}{10} + \frac{2}{11} + \frac{2}{12} \\ &= \sum_{r=1}^4 \frac{2}{8+r} \end{aligned}$$

$$\begin{aligned}
\text{Total area of } n \text{ rectangles} &= \frac{1}{n} \left[\frac{2}{1 + \left(1 + \frac{1}{n}\right)} + \frac{2}{1 + \left(1 + \frac{2}{n}\right)} + \dots + \frac{2}{1 + \left(1 + \frac{n-1}{n}\right)} + \frac{2}{1 + 2} \right] \\
&= \frac{1}{n} \left[\frac{2}{1 + \left(\frac{n+1}{n}\right)} + \frac{2}{1 + \left(\frac{n+2}{n}\right)} + \dots + \frac{2}{1 + \left(\frac{2n-1}{n}\right)} + \frac{2}{1 + \left(\frac{2n}{n}\right)} \right] \\
&= \frac{1}{n} \left[\frac{2n}{2n+1} + \frac{2n}{2n+2} + \dots + \frac{2n}{2n+n-1} + \frac{2n}{2n+n} \right] \\
&= \frac{2}{2n+1} + \frac{2}{2n+2} + \dots + \frac{2}{2n+n-1} + \frac{2}{2n+n} \\
&= \sum_{r=1}^n \frac{2}{2n+r}
\end{aligned}$$

$$\text{Area under graph} = \int_1^2 \frac{2}{1+x} dx = \left[2 \ln |1+x| \right]_1^2 = 2 \ln 3 - 2 \ln 2 = 2 \ln \frac{3}{2}$$

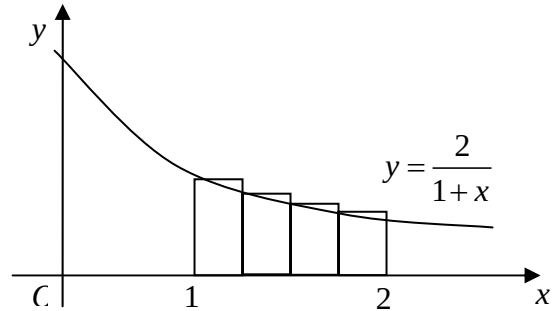
Since Sum of Area of Rectangles < Area under graph

$$\Rightarrow \sum_{r=1}^n \frac{2}{2n+r} < 2 \ln \left(\frac{3}{2} \right) \text{ ----- (1)}$$

Consider rectangles as seen in the diagram,

Total area of n rectangles

$$\begin{aligned}
&= \frac{1}{n} \left[\frac{2}{1+1} + \frac{2}{1 + \left(1 + \frac{1}{n}\right)} + \dots + \frac{2}{1 + \left(1 + \frac{n-1}{n}\right)} \right] \\
&= \frac{1}{n} \left[\frac{2n}{2n} + \frac{2n}{2n+1} + \dots + \frac{2n}{2n+n-1} \right] \\
&= \sum_{r=1}^n \frac{2}{2n+r-1}
\end{aligned}$$



Since Sum of Area of Rectangles > Area under graph

$$\Rightarrow \sum_{r=1}^n \frac{2}{2n+r-1} > 2 \ln \left(\frac{3}{2} \right) \text{ ----- (2)}$$

Considering (1) and (2),

$$\sum_{r=1}^n \frac{2}{2n+r} < 2 \ln \left(\frac{3}{2} \right) < \sum_{r=1}^n \frac{2}{2n+r-1} \text{ . (deduced)}$$

10. Objective: Integration techniques

$$\begin{aligned}
 \text{(a)} \quad \int_{\frac{\pi}{6}}^0 \frac{5 \sin x - 3 \cos x}{\cos x - \sin x} dx &= \int_{\frac{\pi}{6}}^0 \frac{(\cos x + \sin x) - 4(\cos x - \sin x)}{\cos x - \sin x} dx \\
 &= \int_{\frac{\pi}{6}}^0 \frac{\cos x + \sin x}{\cos x - \sin x} - 4 dx \\
 &= \left[-\ln |\cos x - \sin x| - 4x \right]_{\frac{\pi}{6}}^0 \\
 &= \frac{2}{3}\pi + \ln \left(\frac{\sqrt{3}-1}{2} \right)
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad \frac{d}{dx} \left(x(1-x^2)^{\frac{1}{2}} \right) &= \frac{x}{2}(1-x^2)^{-\frac{1}{2}}(-2x) + (1-x^2)^{\frac{1}{2}} \\
 &= \frac{x^2 + (1-x^2)}{\sqrt{1-x^2}} \\
 &= \frac{1-2x^2}{\sqrt{1-x^2}} \\
 \rightarrow \int \frac{1-2x^2}{\sqrt{1-x^2}} dx &= \left(x(1-x^2)^{\frac{1}{2}} \right) + C \\
 \int_0^{\frac{1}{2}} \frac{3-2x^2}{\sqrt{1-x^2}} - \frac{2}{\sqrt{1-x^2}} dx &= \left[x\sqrt{1-x^2} \right]_0^{\frac{1}{2}} \\
 \int_0^{\frac{1}{2}} \frac{3-2x^2}{\sqrt{1-x^2}} dx &= \left[x\sqrt{1-x^2} + 2 \sin^{-1} x \right]_0^{\frac{1}{2}} \\
 &= \frac{\pi}{3} + \frac{\sqrt{3}}{4}
 \end{aligned}$$

11. Objective: Inverse function , Composite function, transformation of curve

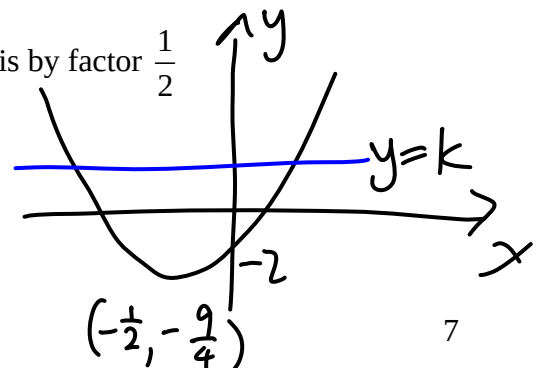
(i) $g(x) = f(x-1)$

The transformation is a translation of +1 unit in the direction of the x-axis

Alternatively, $g(x) = 2^{x-1} = \frac{1}{2}(2^x) = \frac{1}{2}f(x)$

The transformation is a scaling parallel to the y-axis by factor $\frac{1}{2}$

(ii) A horizontal line $y = k$ ($k > -\frac{9}{4}$) cuts the graph at 2 points, h is not one- one , so h^{-1} does not exist



$$(iii) \quad a = -\frac{1}{2}$$

$$h : x \mapsto x^2 + x - 2, \quad x \in \mathbb{R}, x > -\frac{1}{2}$$

$$\text{Let } y = h(x) = x^2 + x - 2, \quad x > -\frac{1}{2}$$

$$y = \left(x + \frac{1}{2}\right)^2 - \frac{9}{4}$$

$$\left(x + \frac{1}{2}\right)^2 = y + \frac{9}{4} = \frac{4y + 9}{4}$$

$$x = -\frac{1}{2} \pm \frac{\sqrt{4y + 9}}{2}$$

$$x = -\frac{1}{2} + \frac{\sqrt{4y + 9}}{2} \quad (\text{since } x > -\frac{1}{2})$$

$$h^{-1} : x \mapsto \frac{-1 + \sqrt{4x + 9}}{2}, \quad x \in \mathbb{R}, x > -\frac{9}{4}$$

$$(iv) \quad R_f = (0, \infty)$$

$$D_{h^{-1}} = \left(-\frac{9}{4}, \infty\right)$$

$$\therefore R_f \subset D_{h^{-1}}.$$

Hence $h^{-1}f$ exists.

$$h^{-1}f(x) = h^{-1}(2^x)$$

$$= \frac{-1 + \sqrt{4(2^x) + 9}}{2}$$

$$h^{-1}f : x \mapsto \frac{-1 + \sqrt{4(2^x) + 9}}{2}, \quad x \in \mathbb{R}$$

$$R_{h^{-1}f} = (1, \infty)$$

12. Objective: Curve sketching, awareness of window settings and equivalence of number of roots to number of points of intersection between graphs

$$x = 5 \text{ is an asymptote} \Rightarrow d = -5$$

$$y = \frac{ax^2 + bx + c}{x - 5} = ax + (b + 5a) + \frac{R}{x - 5}$$

$$y = x + 7 \text{ is an asymptote} \Rightarrow \begin{cases} a = 1, \\ b + 5a = 7 \Rightarrow b = 2 \end{cases}$$

$$y = \frac{x^2 + 2x + c}{x - 5} \Rightarrow \frac{dy}{dx} = \frac{(x-5)(2x+2) - (x^2 + 2x + c)}{(x-5)^2}$$

$$\text{At } x = -1, \frac{dy}{dx} = 0 \Rightarrow (-1-5)(-2+2) - (1-2+c) = 0 \Rightarrow c = 1$$

$$\text{Sketch } y = \frac{x^2 + 2x + 1}{x - 5}$$

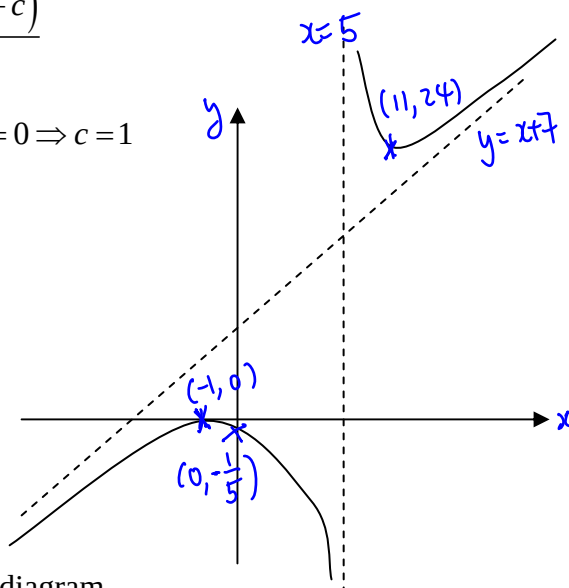
$$(1-\lambda)x^2 + (2+5\lambda)x + 1 = 0$$

$$\Rightarrow x^2 - \lambda x^2 + 2x + 5\lambda x + 1 = 0$$

$$\Rightarrow x^2 + 2x + 1 = \lambda x^2 - 5\lambda x$$

$$\Rightarrow \frac{x^2 + 2x + 1}{x - 5} = \lambda x$$

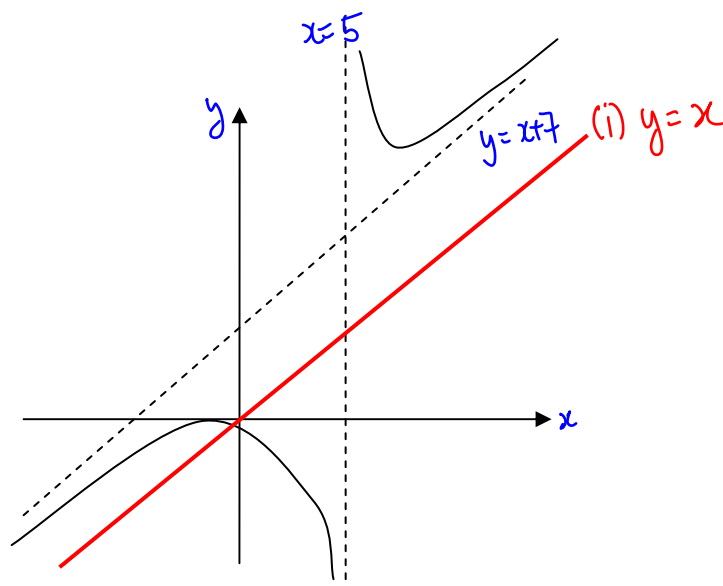
Therefore, sketch the graph of $y = \lambda x$ on the same diagram.



(i) $y = \lambda x, \lambda = 1$

From graph, only one pt of intersection

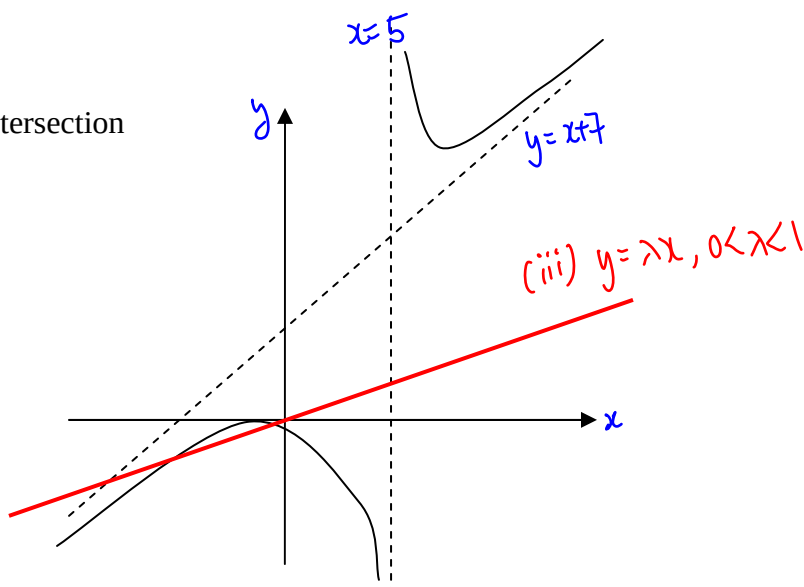
$\Rightarrow$ one real root



(ii) $y = \lambda x, 0 < \lambda < 1$

From graph, two pts of intersection

$\Rightarrow$ two real roots



13(a) Objective: Application of DE, First & Second ODE

Let x be the amount of radium at any time t .

$$\frac{dx}{dt} = -kx$$

$$\int \frac{1}{x} dx = -k \int dt$$

$$\ln x = -kt + c$$

If m is the initial amount, then $\ln m = c$

$$x = \frac{1}{2}m \text{ when } t = 1600$$

$$\ln \frac{m}{2} = -1600k + \ln m$$

$$k = \frac{\ln 2}{1600}$$

$$\ln x = -kt + c$$

When $x = \frac{99}{100}m$, $\ln \frac{99m}{100} = -\frac{\ln 2}{1600}t + \ln m$, $t = 23.2$

It will take 23 years to lose 1% of the initial mass.

13(b)
$$\frac{d^2x}{dt^2} = \frac{1}{1+4t^2} = \frac{1}{4\left(\frac{1}{4}+t^2\right)}$$

$$\begin{aligned} \frac{dx}{dt} &= \frac{1}{4} \int \frac{1}{\left(\frac{1}{4}+t^2\right)} dt \\ &= \left(\frac{1}{4}\right) \left(\frac{1}{1/2}\right) \tan^{-1} \frac{t}{1/2} + c \\ &= \frac{1}{2} \tan^{-1} 2t + c \end{aligned}$$

$$\frac{dx}{dt} = 0 \text{ when } t = 0 \Rightarrow c = 0$$

$$\frac{dx}{dt} = \frac{1}{2} \tan^{-1} 2t$$

$$\begin{aligned} x &= \frac{1}{2} \int \tan^{-1} 2t dt \\ &= \frac{1}{2} \left[t \tan^{-1} 2t - \int \frac{2t}{1+4t^2} dt \right] \\ &= \frac{1}{2} \left[t \tan^{-1} 2t - \frac{1}{4} \ln(1+4t^2) + d \right] \end{aligned}$$

$$x = 0.5 \text{ when } t = 0 \Rightarrow d = 1 \quad \therefore x = \frac{1}{8} [4t \tan^{-1} 2t - 2 \ln(1+4t^2) + 4]$$